

National University of Sciences & Technology
School of Electrical Engineering and Computer Science
Department of Humanities and Sciences

MATH-234: Multivariable Calculus (3+0): 2k25-BSDS-3A, 2k25-BSAI-2A- Spring 2026

Assignment 1	
CLO-1: Understand the concept of limit, continuity and derivatives.	
Announcement Date: 23 Feb 2026	Due Date: 2 March 2026

Instructions:

1. This is an individual assignment.
2. Use this page as cover page (otherwise marks will be deducted).
3. Assignment should be hand written. Marks will be deducted if submitted after deadline.
4. CR will collect the assignments and hand over to me.
5. Assignment should be neat and properly stapled. Do not use files and folders.

Tasks: Attempt all questions.

Student Name	QALAM ID	Class/Section

Total Marks	Marks Obtained
10 Marks	

Question 1:

For the following functions

- I. $f(x, y) = \ln(1 - x + y) + \sqrt{9 - x^2 - y^2}$
- II. $f(x, y) = \sqrt{(x + y)(2 - x)}$
 - a. Determine domain(Show working), explain it and represent it geometrically.
 - b. State whether it is open, closed, neither, bonded or unbounded.

Question 2:

- I. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{1+x^2+y^2}-1}{x^2+y^2}$
- II. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2-y^2}{x^2+y^2}$
 - a. Along coordinate axis.
 - b. Along the lines passing through origin.
 - c. Along the parabolas $y = mx^2$.
 - d. Conclude whether the limit exists?
- III. A temperature field on a surface is given as
$$T(x, y) = \frac{xy}{x^2 + y^2}$$
 - a. Determine whether the temperature approaches a finite value at the origin.
 - b. Determine whether the temperature is direction dependent near origin.

Question 3:

Consider the surfaces

- I. $y^2 = x^2 + z^2$
- II. $x^2 - y^2 - z^2 = 1$
 - a. Identify the type of surface by writing it in standard form.
 - b. Discuss the symmetries and intercepts of the surface.
 - c. Determine traces in coordinate planes.
 - d. Calculate traces for (I) $y = \pm 1$ and for (II) $x = \pm 3$.
 - e. Sketch the surface. Label the traces and level curves.

Question 4:

On a certain mountain, the elevation z above a point (x, y) in an xy - plane at sea level is $z = 2000 - 0.02x^2 - 0.04y^2$, where x, y , and z are in meters. The positive x - axis points east, and the positive y - axis north. A climber is at the point $(-20, 5, 1991)$.

- a. If the climber uses a compass reading to walk due west, will she begin to ascend or descend?
- b. If the climber uses a compass reading to walk northeast, will she ascend or descend? At what rate?
- c. In what compass direction should the climber begin walking to travel a level path (two answers)?